PHYSICS

Standard Level

Set A | Paper 1A and 1B

Challenge mock examination

Time allowed: 90 minutes Maximum mark: 45

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• Paper 1A and Paper 1B share one uninterrupted examination time. Paper 1A has 25 multiple-choice questions; Paper 1B has two compulsory experimental/data questions worth 20 marks altogether.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

• The number in square brackets gives the maximum mark for a written part.

Thriving Scholars challenge practice | Current Physics syllabus | 2026

Paper 1A | Question paper

Paper 1A - Multiple choice

Select one answer, A, B, C or D, for each question. Use the answer grid or record your choices on the portal.

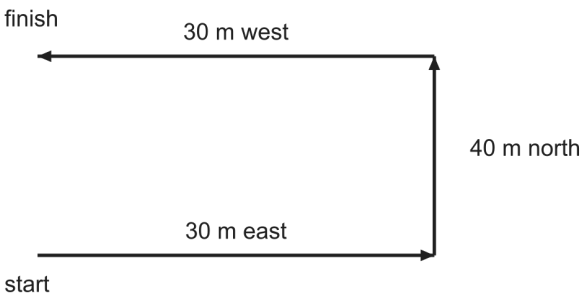
1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20

21	22	23	24	25

1A Question 1 [1 mark]

A surveyor walks the three legs shown in 80 s. What are the average speed and the magnitude of the average velocity, respectively?



A. $0.500\ \mathrm{m\,s^{-1}}$ and $1.25\ \mathrm{m\,s^{-1}}$

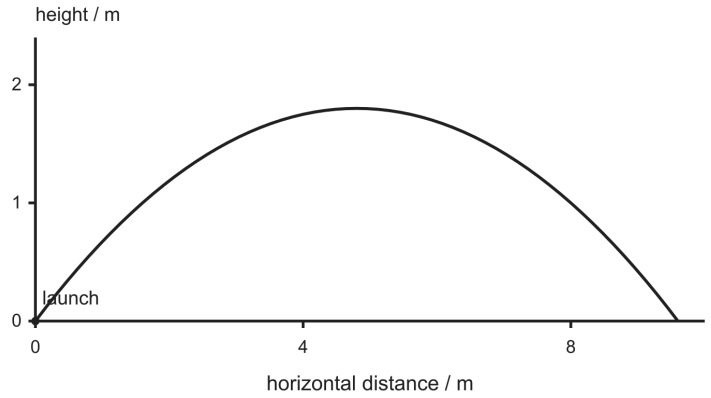
B. $0.875\ \mathrm{m\,s^{-1}}$ and $0.625\ \mathrm{m\,s^{-1}}$

C. $1.25\ \mathrm{m\,s^{-1}}$ and $1.25\ \mathrm{m\,s^{-1}}$

D. $1.25\ \mathrm{m\,s^{-1}}$ and $0.500\ \mathrm{m\,s^{-1}}$

1A Question 2 [1 mark]

A projectile is launched with horizontal and upward velocity components $8.0\,\mathrm{m\,s^{-1}}$ and $6.0\,\mathrm{m\,s^{-1}}$. Air resistance is negligible; use $g = 10\,\mathrm{m\,s^{-2}}$. At what horizontal distance is it at height $1.6\,\mathrm{m}$ while descending?



A. $6.4\,\mathrm{m}$

B. $9.6\,\mathrm{m}$

C. $4.8\,\mathrm{m}$

D. $3.2\,\mathrm{m}$

1A Question 3 [1 mark]

Two identical balls in separate trials pass a chosen height with equal speeds, one moving upward and the other downward. At this speed each ball has drag magnitude $0.40mg$. What is the ratio of the upward-moving ball's acceleration magnitude to the downward-moving ball's acceleration magnitude?

A. $5/2$

B. $3/7$

C. $7/3$

D. 1

1A Question 4 [1 mark]

On a horizontal frictionless track, cart X ($2.0\,\text{kg}$) moves right at $6.0\,\text{m}\,\text{s}^{-1}$ while cart Y ($1.0\,\text{kg}$) moves left at $3.0\,\text{m}\,\text{s}^{-1}$. After a collision Y moves right at $5.0\,\text{m}\,\text{s}^{-1}$. How much total kinetic energy is dissipated?



A. $40.5\,\text{J}$

B. $24\,\text{J}$

C. $16\,\text{J}$

D. $18\,\text{J}$

1A Question 5 [1 mark]

A spring obeys Hooke's law. Stretching it from extension $0.040\,\text{m}$ to $0.100\,\text{m}$ requires $0.84\,\text{J}$. What energy is stored at extension $0.080\,\text{m}$, measured from its natural length?

A. $1.12\,\text{J}$

B. $1.28\,\text{J}$

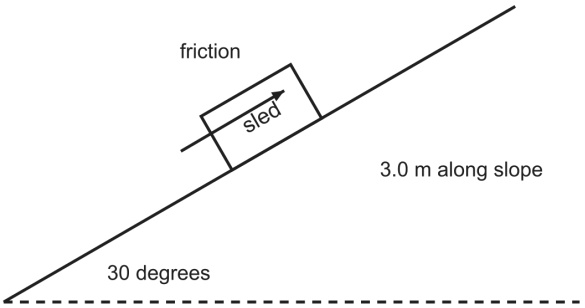
C. $0.64\,\text{J}$

D. $0.56\,\text{J}$

Paper 1A | Question paper

1A Question 6 [1 mark]

A $2.0\,\text{kg}$ sled starts from rest and slides $3.0\,\text{m}$ down a slope at 30° to the horizontal. The constant friction force is $3.0\,\text{N}$. Use $g = 10\,\text{m}\,\text{s}^{-2}$. What is its final speed?



A. $4.6\,\text{m}\,\text{s}^{-1}$

B. $6.2\,\text{m}\,\text{s}^{-1}$

C. $5.5\,\text{m}\,\text{s}^{-1}$

D. $3.0\,\text{m}\,\text{s}^{-1}$

1A Question 7 [1 mark]

An insulated container mixes $0.30\,\text{kg}$ of liquid P at 10°C with $0.20\,\text{kg}$ of liquid Q at 70°C. The final temperature is 30°C. Heat absorbed by the container is negligible. What is c_P/c_Q?

A. $3/4$

B. 2

C. $1/2$

D. $4/3$

1A Question 8 [1 mark]

Surfaces X and Y have the same temperature. X receives $400\,\text{W}\,\text{m}^{-2}$, reflects $80\,\text{W}\,\text{m}^{-2}$ and emits $300\,\text{W}\,\text{m}^{-2}$ as infrared. Y receives $600\,\text{W}\,\text{m}^{-2}$, reflects $180\,\text{W}\,\text{m}^{-2}$ and emits $200\,\text{W}\,\text{m}^{-2}$. Which statement follows?

A. X has the greater albedo and $\varepsilon_X/\varepsilon_Y = 2/3$.

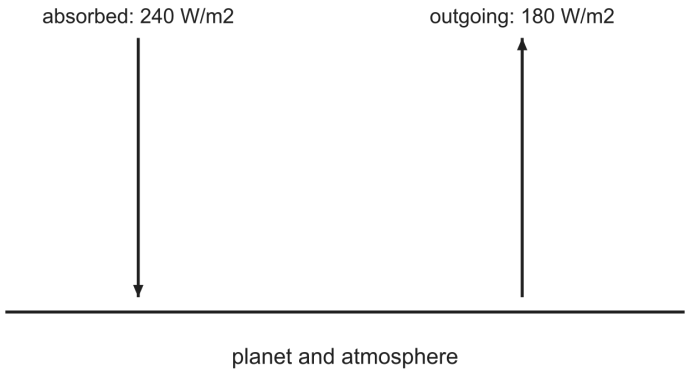
B. Y has the greater albedo and $\varepsilon_X/\varepsilon_Y = 2/3$.

C. Y has the greater albedo and $\varepsilon_X/\varepsilon_Y = 1.50$.

D. X has the greater albedo and $\varepsilon_X/\varepsilon_Y = 1.50$.

1A Question 9 [1 mark]

A planet initially absorbs and emits $240\ \mathrm{W\,m^{-2}}$ at the top of its atmosphere. An increase in infrared-absorbing gas initially reduces the outgoing intensity to $180\ \mathrm{W\,m^{-2}}$, without changing absorbed sunlight. Which prediction is consistent with energy conservation?



A. The gas permanently removes $60\ \mathrm{W\,m^{-2}}$ of energy from the universe.

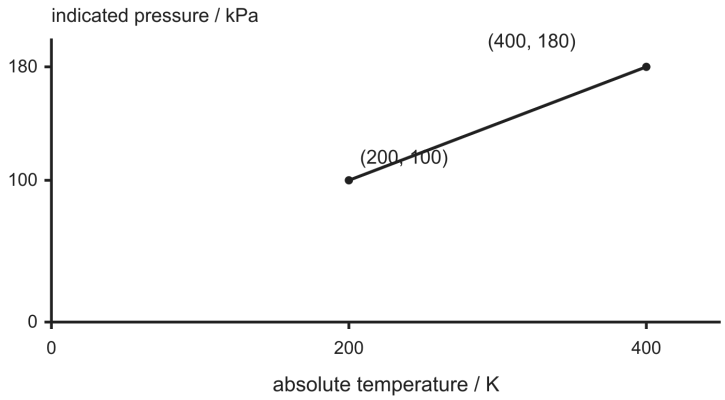
B. The planet initially loses energy and cools until it emits $180\ \mathrm{W\,m^{-2}}$.

C. The planet remains at the same temperature because incoming sunlight is unchanged.

D. The planet initially gains energy, and warming can restore outgoing intensity to $240\ \mathrm{W\,m^{-2}}$.

1A Question 10 [1 mark]

An ideal gas occupies a rigid $0.050\ \mathrm{m^3}$ vessel. Its indicated pressure follows the plotted straight line. The pressure sensor has a constant zero error. Use $R = 8.31\ \mathrm{J\,mol^{-1}\,K^{-1}}$. How many moles of gas are present?



A. $2.41\ \mathrm{mol}$

B. $1.66\ \mathrm{mol}$

C. $0.00241\ \mathrm{mol}$

D. $3.61\ \mathrm{mol}$

1A Question 11 [1 mark]

Two monatomic ideal-gas samples have the same temperature. Sample X contains twice as many atoms as sample Y, and each X atom has one-fifth the mass of a Y atom. What are U_X/U_Y and $v_{\mathrm{rms,X}}/v_{\mathrm{rms,Y}}$?

A. 1 and $\sqrt{5}$

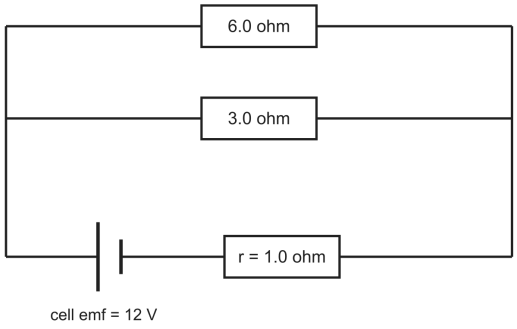
B. 2 and $\sqrt{5}$

C. 10 and $1/\sqrt{5}$

D. 2 and 5

1A Question 12 [1 mark]

A $12\,\mathrm{V}$ cell with internal resistance $1.0\,\Omega$ supplies the parallel network shown. What fraction of the cell's electrical power is dissipated in the external resistors?



A. $1/3$

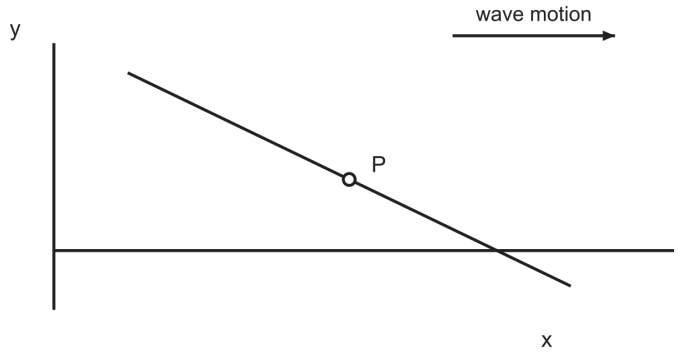
B. 1

C. $2/3$

D. $8/9$

1A Question 13 [1 mark]

A transverse wave travels in the positive x direction at $5.0\,\mathrm{m\,s^{-1}}$. At point P the instantaneous spatial gradient $\Delta y/\Delta x$ is -0.080. For a small time interval, the wave profile translates without changing shape. What is the instantaneous transverse velocity of P?



A. $0.016\,\mathrm{m\,s^{-1}}$ upward

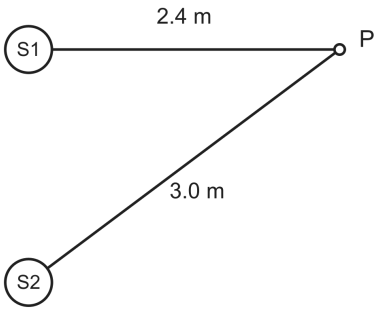
B. $0.40\,\mathrm{m\,s^{-1}}$ upward

C. $5.0\,\mathrm{m\,s^{-1}}$ upward

D. $0.40\,\mathrm{m\,s^{-1}}$ downward

1A Question 14 [1 mark]

Two coherent loudspeakers initially emit in phase at wavelength $0.40\,\mathrm{m}$. Point P is $2.4\,\mathrm{m}$ from one speaker and $3.0\,\mathrm{m}$ from the other; the arriving amplitudes are equal. One speaker is then shifted by half a cycle. What happens at P?



A. Destructive interference changes to constructive interference.

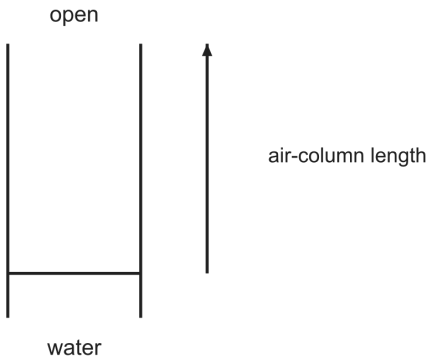
B. It stays destructive because the distances do not change.

C. Interference disappears because the frequencies are now different.

D. Constructive interference changes to destructive interference.

1A Question 15 [1 mark]

A tube is closed by a movable water surface. Its first two resonance lengths for a tuning fork are $0.180\,\text{m}$ and $0.540\,\text{m}$. The end correction is the same at both resonances. The speed of sound is $343\,\text{m}\,\text{s}^{-1}$. What is the fork frequency?



A. $635\,\text{Hz}$

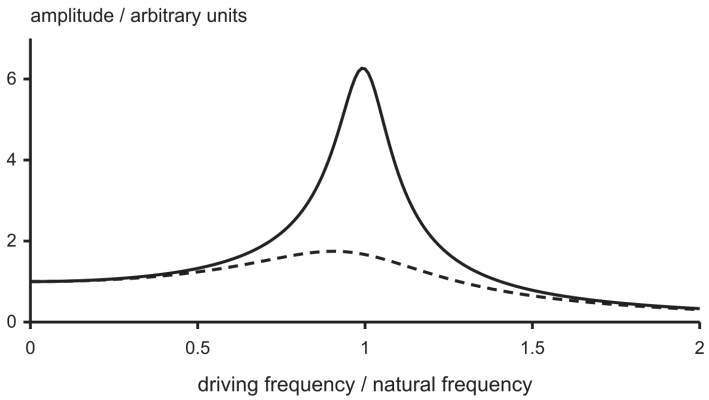
B. $476\,\text{Hz}$

C. $953\,\text{Hz}$

D. $318\,\text{Hz}$

1A Question 16 [1 mark]

The solid and dashed curves show the steady-state displacement amplitude of the same driven oscillator under two damping conditions. The driving-force amplitude is unchanged. Which conclusion is supported?



A. The dashed response has greater damping and a broader, lower resonance peak.

B. The dashed response has less damping and a broader, lower resonance peak.

C. The solid response has greater damping because its peak is taller.

D. The natural frequency must double when damping is increased.

1A Question 17 [1 mark]

A planet follows a circular orbit of radius r and period T about star X. Another planet orbits star Y with radius $2r$ and period $T/2$. Planet masses are negligible. What is M_Y/M_X?

A. 8

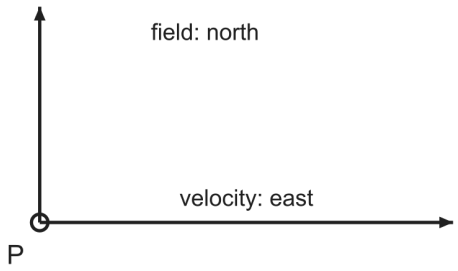
B. 32

C. 2

D. $1/2$

1A Question 18 [1 mark]

At P a probe has velocity due east and gravitational field strength due north. Gravity is its only force. What are its acceleration direction and its instantaneous rate of change of speed at P?



A. North; positive

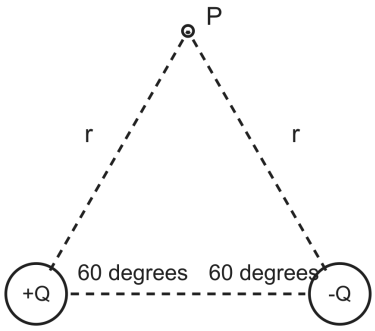
B. East; zero

C. North; zero

D. East; positive

1A Question 19 [1 mark]

Two charges $+Q$ and $-Q$ are separated horizontally. Point P is distance r from each charge, with the lines to P making 60° to the horizontal. Let $E_0 = kQ/r^2$. What is the resultant electric field at P?



A. Zero

B. $2E_0$ to the right

C. $\sqrt{3}E_0$ upward

D. E_0 to the right

1A Question 20 [1 mark]

A particle of mass m and positive charge q passes undeflected through crossed fields, then enters a region containing only the same magnetic field, following radius r. A second particle has mass $2m$ and charge $q/2$. It also passes undeflected through the selector. What is its analyser radius?

A. $r/4$

B. $2r$

C. r

D. $4r$

1A Question 21 [1 mark]

A very thin metal foil produces a small number of large-angle alpha-particle deflections. Which observation most directly supports concentration of positive charge in a very small nucleus?

A. All incident particles turn back from the foil surface.

B. A tiny fraction of incident particles reverse direction, while most pass nearly straight through.

C. Every incident particle loses exactly the same fraction of its energy.

D. The scattered particles have a continuous range of negative charges.

1A Question 22 [1 mark]

A uranium nucleus $^{238}_{92}\mathrm{U}$ undergoes alpha decay. Its daughter subsequently undergoes beta-minus decay. What are the final nucleon number and proton number?

A. 238 and 93

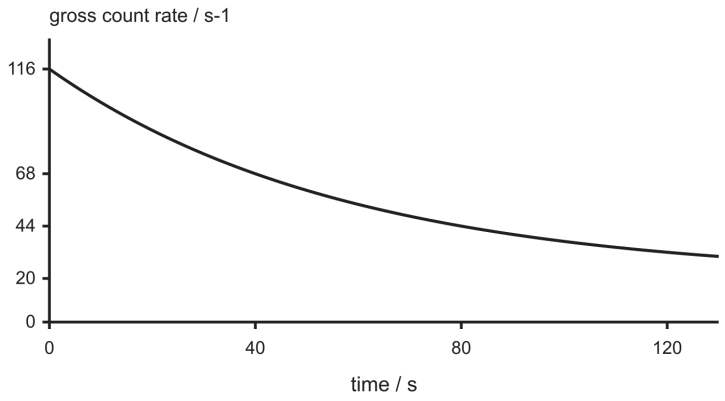
B. 234 and 89

C. 234 and 91

D. 236 and 90

1A Question 23 [1 mark]

A source has half-life 40 s. A detector measures $116\,\mathrm{s}^{-1}$ at $t = 0$ and $44\,\mathrm{s}^{-1}$ at $t = 80\,\mathrm{s}$. Background is constant. What is the ratio of source count rate to background count rate at $t = 120\,\mathrm{s}$?



A. 0.40

B. 1.20

C. 0.145

D. 0.60

1A Question 24 [1 mark]

A uranium-235 nucleus captures one neutron and splits into nuclei with nucleon numbers 141 and 92, plus free neutrons. Binding energy per nucleon is $7.6\,\text{MeV}$ for uranium, $8.4\,\text{MeV}$ for the first fragment and $8.6\,\text{MeV}$ for the second. Approximately how much energy is released?

A. $190\,\text{MeV}$

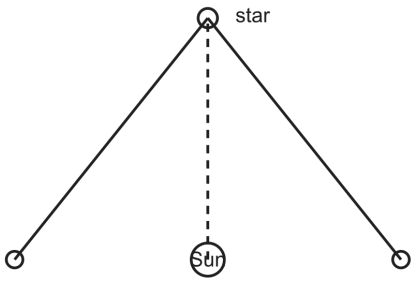
B. $7.8\,\text{MeV}$

C. $1980\,\text{MeV}$

D. $330\,\text{MeV}$

1A Question 25 [1 mark]

A nearby star's greatest apparent angular displacement between observations six months apart is 0.100 arc-seconds. Proper motion has been corrected. Use $1\,\text{pc} = 3.26\,\text{ly}$. What is its distance?



opposite Earth positions, six months apart

A. $130\,\text{ly}$

B. $6.52\,\text{ly}$

C. $65.2\,\text{ly}$

D. $32.6\,\text{ly}$

1B Question 1 | Question paper

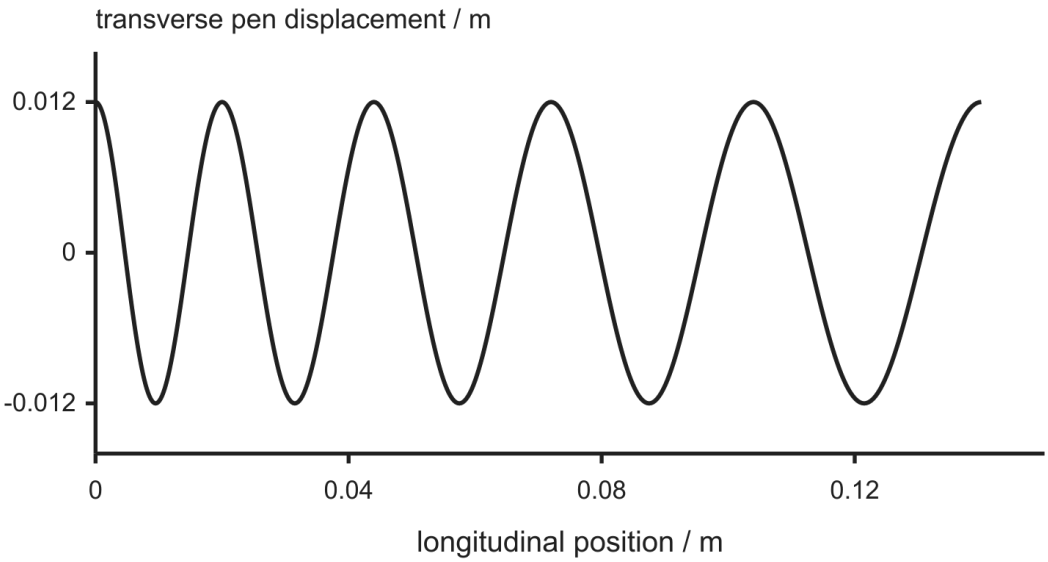
Paper 1B - Experimental and data-based questions

Continue within the same examination time. Answer both questions. Use the data, graphs and uncertainty information provided.

1B Question 1 [10 marks]

A pen attached to a cart oscillates perpendicular to the direction of travel. Its calibrated frequency is $(10.0 \pm 0.2)\,\text{Hz}$. The trace records successive positive peaks at the longitudinal positions below. Each position has maximum uncertainty $0.001\,\text{m}$. Treat the frequency uncertainty as a maximum uncertainty and use a constant-acceleration model.

Peak number	0	1	2	3	4	5
Position / m	0.000	0.020	0.044	0.072	0.104	0.140



(a) Calculate the time between successive peaks and its maximum uncertainty. [2]

(b) Estimate the longitudinal velocity at peak 2 using peaks 1 and 3. [2]

(c) Use the change in successive peak spacings to determine the acceleration. [2]

(d) Calculate the maximum uncertainty in your velocity estimate in (b). [2]

(e) A student says that increasing longitudinal peak spacings prove the pen frequency is increasing. Evaluate this claim. [2]

1B Question 2 | Question paper

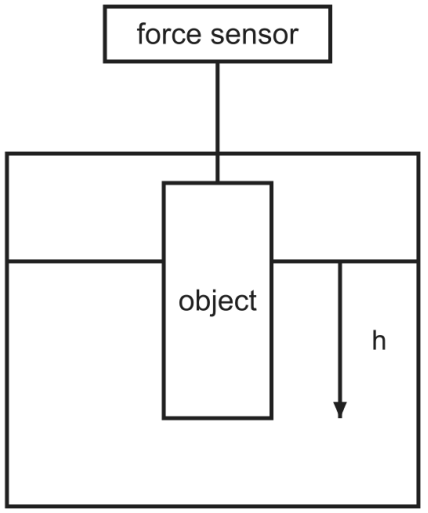
1B Question 2 [10 marks]

A force sensor suspends a uniform vertical object partly in a liquid. The cross-sectional area is $(4.00 \pm 0.08) \times 10^{-4}\,\mathrm{m}^2$. Immersion depth h remains below the object's total length. No part touches the vessel, surface-tension effects are negligible and the object remains stationary. The measured upward sensor force is F. A fit to the laboratory data gives

$$F = (2.35 \pm 0.02) - (3.90 \pm 0.15)h,$$

with F in newtons and h in metres. Use $g = 9.81\,\mathrm{m\,s^{-2}}$.

h/m	0.00	0.03	0.06	0.09	0.12
F/N	2.35	2.23	2.12	2.00	1.88



stationary object

uniform cross-section

no base contact

(a) Identify the forces on the object and derive $F = W - \rho g A h$. [2]

(b) Use the gradient to determine the liquid density. [3]

(c) Determine the maximum uncertainty in that density. Neglect uncertainty in g. [2]

1B Question 2 | Question paper

(d) Would a constant $+0.10\,\text{N}$ force-sensor zero error change the density obtained by this method? Explain. [1]

(e) Explain why the straight-line model must stop applying once the object becomes completely immersed. [2]

End of question paper

PHYSICS

Standard Level

Set A | Paper 1A and 1B

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 1A - Answer key and explanations

1	2	3	4	5	6	7	8	9	10
D	A	C	B	C	A	D	C	D	A

11	12	13	14	15	16	17	18	19	20
B	C	B	A	B	A	B	C	D	D

21	22	23	24	25
B	C	D	A	C

1A Question 1 [1 mark]

Correct answer: D

The distance is $30 + 40 + 30 = 100\,\text{m}$, whereas the displacement is $40\,\text{m}$ north. Divide each by $80\,\text{s}$. The order of the requested quantities matters. Therefore choose D.

1A Question 2 [1 mark]

Correct answer: A

Solve $1.6 = 6t - 5t^2$ to obtain $t = 0.40$ or $0.80\,\text{s}$. Since $v_y = 6 - 10t$, descent occurs at $0.80\,\text{s}$. Thus $x = 8(0.80) = 6.4\,\text{m}$. The smaller root is the upward crossing. Therefore choose A.

1A Question 3 [1 mark]

Correct answer: C

On ascent, weight and drag both act down, giving $1.40g$. On descent, drag acts up, giving $0.60g$ down. The ratio is $1.40/0.60 = 7/3$. Equal speeds give equal drag magnitudes, not equal resultant forces. Therefore choose C.

1A Question 4 [1 mark]

Correct answer: B

Taking right positive, $2(6) + 1(-3) = 2v_X + 1(5)$, so $v_X = 2.0\,\text{m}\,\text{s}^{-1}$. Initial kinetic energy is $40.5\,\text{J}$ and final is $16.5\,\text{J}$, giving $24\,\text{J}$ dissipated. Momentum needs signed velocities; kinetic energy needs squared speeds. Therefore choose B.

1A Question 5 [1 mark]

Correct answer: C

Use $0.84 = \frac{1}{2}k(0.100^2 - 0.040^2)$, giving $k = 200\,\mathrm{N\,m^{-1}}$. Then $E = \frac{1}{2}(200)(0.080)^2 = 0.64\,\mathrm{J}$. Work between extensions is a difference of squares, not the square of their difference. Therefore choose C.

1A Question 6 [1 mark]

Correct answer: A

The vertical drop is $3\sin 30^\circ = 1.5\,\mathrm{m}$. The kinetic-energy gain is $mg(1.5) - F(3) = 30 - 9 = 21\,\mathrm{J}$. Since $\frac{1}{2}mv^2 = 21$, $v = \sqrt{21} = 4.58\,\mathrm{m\,s^{-1}}$. Therefore choose A.

1A Question 7 [1 mark]

Correct answer: D

Energy conservation gives $0.30c_P(30 - 10) = 0.20c_Q(70 - 30)$. Thus $c_P/c_Q = 8/6 = 4/3$. The different masses as well as the temperature changes must be included. Therefore choose D.

1A Question 8 [1 mark]

Correct answer: C

Albedo is reflected divided by incident intensity: 0.20 for X and 0.30 for Y. At equal temperature, emitted intensity is proportional to emissivity, so $\varepsilon_X/\varepsilon_Y = 300/200 = 1.50$. These data need not describe thermal equilibrium. Therefore choose C.

1A Question 9 [1 mark]

Correct answer: D

Initially the gain is $240 - 180 = 60\,\mathrm{W\,m^{-2}}$. Infrared absorption and re-emission redistribute energy; they do not destroy it. A higher planetary temperature can increase emission until the top-of-atmosphere balance is restored. Therefore choose D.

1A Question 10 [1 mark]

Correct answer: A

For true pressure, $P = nRT/V$. A constant offset changes the intercept but not the gradient. From the plotted points, $\Delta P/\Delta T = (180 - 100)/200 = 0.400\,\mathrm{kPa\,K^{-1}} = 400\,\mathrm{Pa\,K^{-1}}$. Thus $n = (400)(0.050)/8.31 = 2.41\,\mathrm{mol}$. Therefore choose A.

1A Question 11 [1 mark]

Correct answer: B

For a monatomic ideal gas $U = \frac{3}{2}Nk_BT$, so the energy ratio is 2. Also $\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}k_BT$, so the rms speed ratio is $\sqrt{m_Y/m_X} = \sqrt{5}$. Atom number does not set the rms speed at fixed temperature. Therefore choose B.

1A Question 12 [1 mark]

Correct answer: C

The external resistance is $(1/6 + 1/3)^{-1} = 2.0\,\Omega$. The total is $3.0\,\Omega$, so $I = 4.0\,\text{A}$. Useful power is $I^2(2) = 32\,\text{W}$ and cell power is $\varepsilon I = 48\,\text{W}$. The fraction is $2/3$. The same current traverses the internal resistance. Therefore choose C.

1A Question 13 [1 mark]

Correct answer: B

During Δt, the profile moves right by $v\Delta t$. At fixed position P, $\Delta y \approx -(\Delta y/\Delta x)v\Delta t$, so the transverse velocity is $-5.0(-0.080) = +0.40\,\text{m s}^{-1}$. A negative spatial slope in a right-moving wave means upward particle motion. Therefore choose B.

1A Question 14 [1 mark]

Correct answer: A

The path difference is $0.60\,\text{m} = 1.5\lambda$, so initially the waves arrive half a cycle out of phase. An additional half-cycle source shift makes the arrival phase difference a whole number of cycles, giving constructive interference. A fixed phase shift does not change frequency or coherence. Therefore choose A.

1A Question 15 [1 mark]

Correct answer: B

Successive allowed modes of an open-closed tube differ in length by $\lambda/2$. Hence $\lambda = 2(0.540 - 0.180) = 0.720\,\text{m}$ and $f = 343/0.720 = 476\,\text{Hz}$. Subtraction cancels the constant end correction. Therefore choose B.

1A Question 16 [1 mark]

Correct answer: A

Increased damping dissipates more energy per cycle and reduces the build-up of amplitude near resonance. Its response is lower and broader. A taller peak is a sign of less damping, and the data do not imply a doubled natural frequency. Therefore choose A.

1A Question 17 [1 mark]

Correct answer: B

From $T^2 = 4\pi^2 r^3/(GM)$, stellar mass is proportional to r^3/T^2. The ratio is $2^3/(1/2)^2 = 32$. The two planets orbit different stars, so one cannot use a constant T^2/r^3 for both without accounting for mass. Therefore choose B.

1A Question 18 [1 mark]

Correct answer: C

The acceleration is in the field direction, north. At this instant it is perpendicular to velocity, so it changes the direction of motion and the instantaneous tangential acceleration is zero. This does not claim the speed stays constant later. Therefore choose C.

1A Question 19 [1 mark]

Correct answer: D

At P the field from $+Q$ is upward-right and the field toward $-Q$ is downward-right. The vertical components cancel. The horizontal components add to $2E_0 \cos 60^\circ = E_0$ right. Electric field is a vector. Therefore choose D.

1A Question 20 [1 mark]

Correct answer: D

Undeflected motion requires $qE = qvB$, so both selected particles have $v = E/B$, independent of mass and charge. In the magnetic-only region $r = mv/(qB)$. The second ratio is $(2m)/(q/2)$ divided by m/q, giving $4r$. Therefore choose D.

1A Question 21 [1 mark]

Correct answer: B

Rare large-angle/backward deflections require a highly concentrated repulsive region. The near-straight passage of most particles implies that this region occupies a small fraction of atomic volume. Uniform positive charge cannot readily account for the rare strong deflections. Therefore choose B.

1A Question 22 [1 mark]

Correct answer: C

Alpha decay reduces nucleon number by 4 and proton number by 2, yielding $A = 234$, $Z = 90$. Beta-minus changes a neutron into a proton, leaving A unchanged and increasing Z to 91. The emitted electron is not a pre-existing orbital electron. Therefore choose C.

1A Question 23 [1 mark]

Correct answer: D

Let background be b. Two half-lives give $44 - b = (116 - b)/4$, hence $b = 20\,\mathrm{s}^{-1}$. After three half-lives the source contributes $(116 - 20)/8 = 12\,\mathrm{s}^{-1}$. The requested ratio is $12/20 = 0.60$, using source rather than gross rate. Therefore choose D.

1A Question 24 [1 mark]

Correct answer: A

Free neutrons have zero nuclear binding energy. Total final binding is $141(8.4) + 92(8.6) = 1975.6\,\mathrm{MeV}$; initial binding is $235(7.6) = 1786.0\,\mathrm{MeV}$. The increase, $189.6\,\mathrm{MeV}$, is released. Conservation also gives three free neutrons. Therefore choose A.

1A Question 25 [1 mark]

Correct answer: C

The full six-month angular displacement is twice the parallax angle. Thus $p = 0.0500$ arc-seconds, $d = 1/p = 20.0\,\mathrm{pc}$, and $d = 65.2\,\mathrm{ly}$. Using the full displacement as p produces a factor-of-two error. Therefore choose C.

Paper 1B - Worked solutions

1B Question 1 [10 marks]

(a) [2 marks]

M1 $T = 1/f = 0.100\,\text{s}$.

A1 $\delta T/T = \delta f/f = 0.020$, giving $\delta T = 0.002\,\text{s}$.

(b) [2 marks]

M1 Use the central interval: $v \approx (x_3 - x_1)/(2T)$.

A1 $v = (0.072 - 0.020)/0.200 = 0.260\,\text{m s}^{-1}$. For constant acceleration, this is the instantaneous velocity at the midpoint time.

(c) [2 marks]

M1 Successive spacings increase by $0.004\,\text{m}$. With equal time intervals, $\Delta(\Delta x) = aT^2$.

A1 $a = 0.004/(0.100)^2 = 0.400\,\text{m s}^{-2}$.

(d) [2 marks]

M1 $\delta(\Delta x) = 0.002\,\text{m}$ and $\delta(2T) = 0.004\,\text{s}$. Add fractional maximum uncertainties: $\delta v/v = 0.002/0.052 + 0.004/0.200$.

A1 $\delta v = 0.0152\,\text{m s}^{-1}$, so $v = (0.260 \pm 0.015)\,\text{m s}^{-1}$; accept sensible uncertainty rounding such as 0.26 ± 0.02.

(e) [2 marks]

R1 The calibration gives a constant time between peaks; increasing spatial spacing can result from increasing cart speed.

R1 The trace alone does not demonstrate a frequency increase. A separate time calibration is needed; the constant-frequency accelerating-cart model is consistent with the data.

1B Question 2 [10 marks]

(a) [2 marks]

R1 Downward weight W is balanced by upward sensor force and upthrust.

M1 The displaced liquid volume is Ah, so upthrust is $\rho g A h$ and $F + \rho g A h = W$.

(b) [3 marks]

M1 The gradient is $-\rho g A$, with magnitude $3.90\,\mathrm{N\,m^{-1}}$.

M1 $\rho = 3.90/(9.81 \times 4.00 \times 10^{-4})$.

A1 $\rho = 994\,\mathrm{kg\,m^{-3}}$ to three significant figures.

(c) [2 marks]

M1 $\delta\rho/\rho = 0.15/3.90 + 0.08/4.00 = 0.05846$.

A1 $\delta\rho \approx 58\,\mathrm{kg\,m^{-3}}$; accept $(990 \pm 60)\,\mathrm{kg\,m^{-3}}$ or consistent rounding.

(d) [1 mark]

R1 No. A constant force offset changes the intercept, but not the force-depth gradient, provided it is constant throughout the measurements.

(e) [2 marks]

R1 Once fully immersed, the displaced volume stops increasing when the object is lowered farther.

R1 Upthrust and sensor force then remain approximately constant for an incompressible liquid; extending the negative-gradient line would be physically wrong.

End of worked solution book